

Problem Statement - 56 [Economics Part]

⊛ New consumer basket made in 2026

↓
Items increased from ~~244~~ to 233 to 358

↓
Airfares & flight rates added under 'Transport & Communication'

⊛ For CPI calculation → Domestic - T+21
International - T+60

Economic Architecture

Airline website + OTAs

↓
Raw fares

↓
Cleaning & Filtering

↓
Standardised comparable fee

↓
Route x Airline x Booking - window cells

↓
Rate Indices

↓
Weighted Aggregation

↓
National APIA

↓
CPI Air fare item

↓
Overall CPI

What fares to choose?

* 2 important data needed:-

multiple - advance purchase windows

- T+1
- T+7
- T+30
- T+45, etc.

Representative city pairs

- DEL-BOM
- DEL-BLR
- BOM-BLR
- DEL-CCU
- BLR-HYD
- MAA-DEL

DGCA

passenger - carrier data

* Things to build:-

(i) CPI compatible reference fare



For actual CPI calculation

[Domestic = T+21
International = T+60]

(ii) Real Time market index



For the new APIx that would track how airfare changes with booking horizon

[T+1, T+7, T+15, T+45, etc.]

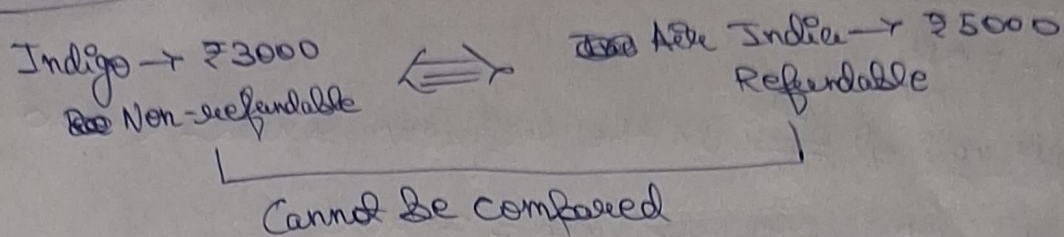
~~Choosing~~

What is fare?

* Only include final mandatory price for the passenger.

Include	Exclude
• Base airfare • mandatory taxes • mandatory airport charges • mandatory surcharge • mandatory fees	• Extra baggage • meals • Insurance • Optional seat allocation, etc.

Standard Ticket

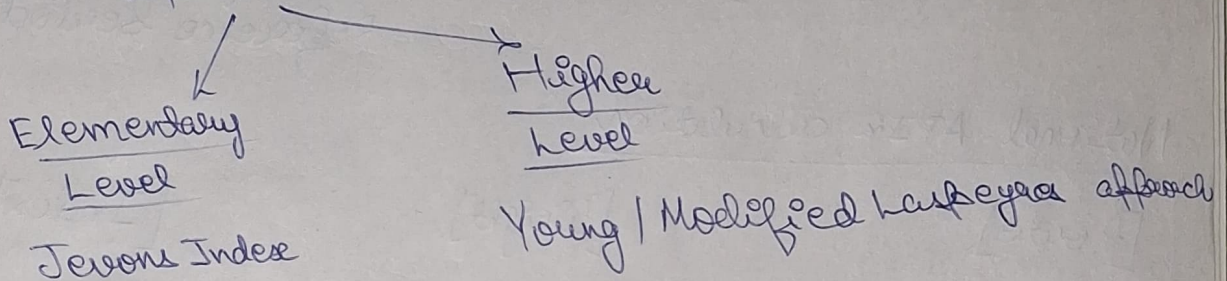


- * So, fare normalization layer needs to be created, such that algorithm compares like with like.

Representative price

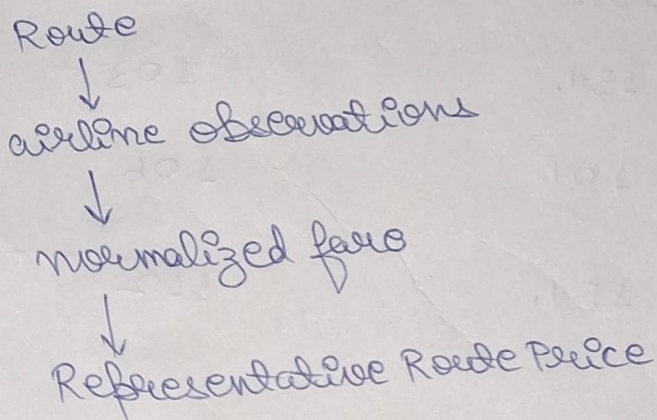
- * Do not use the minimum price.

- * Make a representative basket of all airlines



~~Route~~

- * For Each route:



Actual Index Calculation

Eg :- DEL-BOM

Avg. normalised ^{fare} ~~fare~~ in the reference period = 24000

Today's representative fare = 24400

$$\therefore I_{\text{DEL-BOM}} = \frac{24400}{24000} \times 100 = \boxed{101.67}$$

Airfare is approximately
1.67% higher than the
Reference period

National API Index calculation

[Route]	[Weight]	[Index]
DEL-BOM	25%	101.67
DEL-BLR	20%	101.5
BOM-BLR	15%	101.8
DEL-CCU	15%	101.3
BLR-HYD	10%	101.6
MAA-DEL	15%	101.2

$$API_{\alpha} = \sum_i w_i I_i$$

⊛ Weights come from DGCA passenger-traffic data.

Contribution of airfare in CPI

Eg :- PSD gives weight of airfare in CPI = 0.10%.

4
Airfare index increases by 10%.

$\therefore$ Contribution = $0.10 \times 10\% = 0.01$ percent

$$\text{Contribution} = \text{Weight} \times \text{Inflation}$$

$$= 0.10 \times 10\%$$

$$= 0.01 \text{ percentage points}$$

↓
Airfare contributes ~~0.01~~ 0.01 percentage points to headline inflation

↓
This refers that airfare can increase 10% without CPI increasing 10%.

So 2 things needed -

↓
Airfare Price Index

Eg APIA = 112.4

↓
airfares are ~~above~~ 12.4% above the reference level

↓
CPI contribution

Eg :- Airfare inflation = +12.4%

CPI weight = 0.10%

Estimated CPI = +0.012 percentage points

Dynamic Pricing

Eg :- DEL-BOM [T+1]

8am $\rightarrow$ 25000

12pm $\rightarrow$ 24800

3pm $\rightarrow$ 25600

8pm $\rightarrow$ 25200

$$P_{\text{auto}} = \text{median}(P_1, P_2, \dots, P_n)$$

"Market Index" & "CPI Index"

Real time
Airfare Index

$\rightarrow$ Monitor current
airfare market
conditions

$\rightarrow$ Uses: T+1, T+7,
T+15, T+30,
T+45

$\rightarrow$ High frequency

CPI-compatible
Airfare Index

$\rightarrow$ Provides a statistically
controlled airfare measure
suitable for CPI aggregation

$\rightarrow$ Used: T+21

Economic Methodology

(1) Route ~~and~~ selection

[DCA Passenger - traffic $\rightarrow$ Representative city pairs]



(2) Price collection

[Airline websites + OTAs]



(3) Fare Standardisation

[Economy + standardized passenger + mandatory charges]



(4) Advance purchase windows

$T+1, T+7, T+15, T+30, T+45$

for CPI, $T+21$ (Dom)

$T+60$ (Int)



(5) Data cleaning

(Duplicates, missing values, sold-out, cancellations, outliers)



(6) Elementary Index

[Jevons]



(7) Aggregation [Route weights $\rightarrow$ National API_{air}]



(8) CPI Integration [Airfare Index $\times$ CPI airfare weight]



(9) Policy output [Airfare inflation + CPI contribution]